

## MRC Lecture 4 — Derivation Guide 2

### The Dissociation Constant $K_d$ and Binding Free Energy $\Delta G$

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#### 1. Motivation

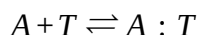
The general Boltzmann distribution (Derivation Guide 1) gives the probability of finding a system in any energy state. For molecular binding, the simplest and most important case is the **two-state system**: a molecule is either bound to its partner or not. Applying the Boltzmann distribution to this two-state system yields the **dissociation constant**  $K_d$  — the quantity measured by every binding assay (SPR, ITC, BLI) for over 150 years.

This guide derives the relationship  $\Delta G = RT \ln K_d$  from the Boltzmann distribution, works through numerical examples spanning the biologically relevant range, and shows that every  $K_d$  measurement is — at its core — a measurement of the Boltzmann probability ratio between bound and unbound states.

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#### 2. The Two-State Binding Equilibrium

Consider an agent molecule A (antibody, enzyme, receptor) and a target molecule T (antigen, substrate, ligand) in solution:



At equilibrium, the forward (association) and reverse (dissociation) rates are equal. The system has two relevant states:

- **Unbound state:** A and T are free in solution. Define the free energy reference at zero:  
 $E_{\text{unbound}} = 0$ .
- **Bound state:** A and T form a complex. Free energy relative to the unbound reference:  
 $E_{\text{bound}} = \Delta G$ .

**Sign convention.** When binding is favorable,  $\Delta G < 0$  (the complex is lower in energy than the separated partners). When binding is unfavorable,  $\Delta G > 0$ . Equilibrium where both states are equally populated corresponds to  $\Delta G = 0$ .

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#### 3. Applying the Boltzmann Distribution

From the Boltzmann distribution (Derivation Guide 1), the probability of each state is:

$$P(\text{bound}) = \frac{\exp(-\Delta G / RT)}{\exp(0) + \exp(-\Delta G / RT)} = \frac{\exp(-\Delta G / RT)}{1 + \exp(-\Delta G / RT)}$$
$$P(\text{unbound}) = \frac{1}{1 + \exp(-\Delta G / RT)}$$

**Gas constant convention.** For molecular applications, we use the gas constant  $R = N_A k_B = 1.987 \times 10^{-3}$  kcal/(mol·K) instead of Boltzmann's constant  $k_B$ , which gives energies in kcal/mol rather than per-molecule units. At physiological temperature  $T = 310$  K:

$$RT = 0.616 \text{ kcal/mol}$$

This is the **thermal energy** — the natural energy scale for molecular recognition.

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#### 4. The Boltzmann Probability Ratio

The **ratio** of bound to unbound probabilities is:

$$\frac{P(\text{bound})}{P(\text{unbound})} = \exp(-\Delta G / RT)$$

This ratio equals the **association constant**  $K_a$ :

$$K_a = \frac{P(\text{bound})}{P(\text{unbound})} = \exp(-\Delta G / RT)$$

The **dissociation constant** is the inverse:

$$K_d = \frac{1}{K_a} = \exp(+\Delta G / RT)$$

Taking the natural logarithm of both sides:

$$\Delta G = RT \ln K_d$$

This is the fundamental equation connecting the measurable  $K_d$  to the thermodynamic quantity  $\Delta G$ . Note that this is **exact** at equilibrium — it is a direct consequence of the Boltzmann distribution, not an approximation.

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## 5. Units and Interpretation of $K_d$

The dissociation constant has units of molar concentration:

$$K_d = \frac{[A][T]}{[A:T]}$$

**Physical interpretation.**  $K_d$  equals the concentration of free target at which half of the agent molecules are bound and half are free. A smaller  $K_d$  means tighter binding: at lower free target concentration, half of the agent is already bound.

**Typical ranges:**

- **Picomolar** ( $K_d \sim 10^{-12}$  M): ultra-tight binding. Biotin-streptavidin is the canonical example; some engineered therapeutic antibodies approach this range.
  - **Nanomolar** ( $K_d \sim 10^{-9}$  M): high-affinity therapeutic antibodies, tight enzyme-substrate interactions.
  - **Micromolar** ( $K_d \sim 10^{-6}$  M): many protein-protein interactions, moderate enzyme-substrate binding.
  - **Millimolar** ( $K_d \sim 10^{-3}$  M): weak, transient interactions typical of signaling and metabolism.
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## 6. Worked Example 1: High-Affinity Antibody

**Problem.** A therapeutic antibody has  $K_d = 10$  nM =  $10^{-8}$  M against its target antigen. Compute  $\Delta G$  at physiological temperature  $T = 310$  K.

**Step 1.** Apply  $\Delta G = RT \ln K_d$ :

$$\Delta G = (0.616 \text{ kcal/mol}) \times \ln(10^{-8})$$

**Step 2.** Compute  $\ln(10^{-8})$ :

$$\ln(10^{-8}) = -8 \times \ln 10 = -8 \times 2.303 = -18.42$$

**Step 3.** Multiply:

$$\Delta G = 0.616 \times (-18.42) = -11.3 \text{ kcal/mol}$$

**Interpretation.** The bound state is 11.3 kcal/mol more stable than the unbound state. In thermal units:  $11.3 / 0.616 \approx 18.3 RT$ . The Boltzmann probability ratio is  $\exp(18.3) \approx 9 \times 10^7$  — at equilibrium with equal free concentrations, the bound state is about 100 million times more probable than the unbound state.

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## 7. Worked Example 2: Moderate-Affinity Interaction

**Problem.** An enzyme has  $K_d = 1 \mu\text{M} = 10^{-6} \text{ M}$  against its substrate. Compute  $\Delta G$  at  $T = 310 \text{ K}$ .

**Step 1.** Apply  $\Delta G = RT \ln K_d$ :

$$\Delta G = (0.616) \times \ln(10^{-6}) = 0.616 \times (-13.82) = -8.5 \text{ kcal/mol}$$

**Interpretation.** In thermal units:  $8.5 / 0.616 \approx 13.8 RT$ . The bound/unbound ratio is  $\exp(13.8) \approx 10^6$  — the bound state is a million times more probable than the unbound state.

**Comparison to Example 1.** The antibody ( $K_d = 10 \text{ nM}$ ) binds about  $100\times$  tighter than the enzyme-substrate pair ( $K_d = 1 \mu\text{M}$ ). In energy: the antibody's  $\Delta G$  is  $2.8 \text{ kcal/mol}$  more negative — about  $4.6 RT$  more favorable.

## 8. The Dynamic Range of Molecular Recognition

The biologically relevant range of binding affinities spans approximately **10 orders of magnitude** in  $K_d$ :

| $K_d$                         | $\Delta G$ at 310 K | In units of $RT$ | Biological examples               |
|-------------------------------|---------------------|------------------|-----------------------------------|
| 1 pM ( $10^{-12} \text{ M}$ ) | -16.5 kcal/mol      | -26.8 $RT$       | Biotin-streptavidin               |
| 10 pM                         | -15.1 kcal/mol      | -24.5 $RT$       | Very high-affinity therapeutics   |
| 100 pM                        | -13.7 kcal/mol      | -22.2 $RT$       | Best therapeutic antibodies       |
| 1 nM                          | -12.7 kcal/mol      | -20.6 $RT$       | Strong antibody-antigen           |
| 10 nM                         | -11.3 kcal/mol      | -18.4 $RT$       | Typical therapeutic antibodies    |
| 100 nM                        | -9.9 kcal/mol       | -16.1 $RT$       | Moderate antibody-antigen         |
| 1 $\mu\text{M}$               | -8.5 kcal/mol       | -13.8 $RT$       | Enzyme-substrate, protein-protein |
| 10 $\mu\text{M}$              | -7.1 kcal/mol       | -11.5 $RT$       | Weak interactions                 |
| 100 $\mu\text{M}$             | -5.7 kcal/mol       | -9.2 $RT$        | Very weak                         |
| 1 mM ( $10^{-3} \text{ M}$ )  | -4.2 kcal/mol       | -6.8 $RT$        | Transient signaling interactions  |

**The entire biologically relevant range:**  $\Delta G$  spans about  $12 \text{ kcal/mol}$ , or  $20 RT$  units.

**What this means quantitatively.** The difference between “tight” ( $K_d = 10 \text{ nM}$ ) and “moderate” ( $K_d = 1 \mu\text{M}$ ) is only  $2.8 \text{ kcal/mol}$  — roughly  $4.6 RT$ . Molecular recognition discriminates between binding partners on energy differences of a few  $RT$ . This is why softmax — not argmax — is the correct selection function: the system is stochastic at this energy scale, and the selection probability is graded, not deterministic.

## 9. Worked Example 3: From Binding Assay to Boltzmann Probability

**Problem.** An SPR experiment measures  $K_d = 500 \text{ pM}$  for an antibody-antigen interaction at  $310 \text{ K}$ .

What is the probability ratio between bound and unbound states when  $[A] = [T]$ ?

**Step 1.** Compute  $\Delta G$ :

$$\Delta G = 0.616 \times \ln(5 \times 10^{-10}) = 0.616 \times (-21.42) = -13.20 \text{ kcal/mol}$$

**Step 2.** Compute the Boltzmann probability ratio:

$$\frac{P(\text{bound})}{P(\text{unbound})} = \exp(-\Delta G / RT) = \exp(13.20 / 0.616) = \exp(21.4) \approx 2 \times 10^9$$

**Interpretation.** At equilibrium with equal free concentrations, the bound state is 2 billion times more probable than the unbound state. The SPR instrument's measurement of  $K_d = 500 \text{ pM}$  IS this Boltzmann probability ratio — expressed in concentration units via the equilibrium constant formalism.

**Generalization.** Every binding measurement ever made — by SPR, ITC, BLI, equilibrium dialysis, fluorescence titration, or any other technique — is a measurement of the ratio  $\exp(-\Delta G / RT)$ . The experimental apparatus differs; the underlying physics does not.

## 10. Converting Between Units

Different fields and textbooks use different units. The relationships are:

**Kelvin, Celsius, and RT:**

| $T$                                | $RT$<br>(kcal/mol) | $RT$<br>(kJ/mol) | $k_B T$<br>(eV) |
|------------------------------------|--------------------|------------------|-----------------|
| 277 K (4°C, cold room)             | 0.550              | 2.303            | 0.0239          |
| 298 K (25°C, standard)             | 0.593              | 2.479            | 0.0257          |
| <b>310 K (37°C, physiological)</b> | <b>0.616</b>       | <b>2.577</b>     | <b>0.0267</b>   |
| 323 K (50°C)                       | 0.642              | 2.687            | 0.0278          |

**Conversions:**

$$\ln(10) \approx 2.303, \text{ so } \Delta G = RT \ln K_d = 2.303 RT \log_{10} K_d$$

At 310 K:  $\Delta G \approx 1.42 \log_{10} K_d$  kcal/mol. For each 10-fold change in  $K_d$ ,  $\Delta G$  changes by about 1.42 kcal/mol (or about 2.3  $RT$ ).

## 11. Connection to the Boltzmann Distribution and Softmax

The two-state  $K_d$  derivation is a special case of the general Boltzmann distribution (Derivation Guide 1, §8):

$$P(j) = \frac{\exp(-E_j / RT)}{\sum_k \exp(-E_k / RT)}$$

With two states ( $j=1$ : unbound,  $E_1=0$ ;  $j=2$ : bound,  $E_2=\Delta G$ ):

$$P(\text{bound}) = \frac{\exp(-\Delta G / RT)}{1 + \exp(-\Delta G / RT)}$$

$$P(\text{unbound}) = \frac{1}{1 + \exp(-\Delta G / RT)}$$

**The partition function** for this two-state system is  $Z = 1 + \exp(-\Delta G / RT)$ .

**Softmax form.** Using the dimensionless score convention  $s_j = -E_j / RT$ :

$$s_1 = 0 \text{ (unbound)}, s_2 = -\Delta G / RT \text{ (bound)}$$

$$P(\text{bound}) = \frac{\exp(s_2)}{\exp(s_1) + \exp(s_2)} = \text{softmax}(s_1, s_2)_2$$

This is the **sigmoid function** in disguise. For a two-state system, softmax reduces to sigmoid:

$$P(\text{bound}) = \sigma(-\Delta G / RT) = \sigma(s)$$

where  $\sigma(x) = 1 / (1 + \exp(-x))$  is the standard sigmoid. However, this is a **special case** — the general molecular recognition problem involves many competitors, where softmax does not reduce to sigmoid. Lecture 6 proves that sigmoid cannot be generalized to a multi-candidate selection function while preserving IIA; only softmax works for the general case. The two-state  $K_d$  derivation is the one place where sigmoid appears naturally, because it is softmax with  $M=2$ .

## 12. Summary

| Concept               | Result                                                       | Significance                              |
|-----------------------|--------------------------------------------------------------|-------------------------------------------|
| Two-state Boltzmann   | $P(\text{bound}) / P(\text{unbound}) = \exp(-\Delta G / RT)$ | Ratio is $K_d$ , the association constant |
| Dissociation constant | $K_d = \exp(+\Delta G / RT) = 1 / K_a$                       | Measured by SPR, ITC, BLI                 |
| Fundamental equation  | $\Delta G = RT \ln K_d$                                      | Exact at equilibrium                      |

| Concept                         | Result                                            | Significance                                  |
|---------------------------------|---------------------------------------------------|-----------------------------------------------|
| Thermal energy at 310 K         | $RT = 0.616$ kcal/mol                             | The natural energy scale                      |
| Biologically relevant range     | $K_d$ : pM to mM; $\Delta G$ : -4 to -16 kcal/mol | Spans $\sim 20 RT$                            |
| Two-state $\rightarrow$ sigmoid | Softmax with $M=2$ = sigmoid                      | Special case; breaks down for multi-candidate |

### 13. Checkpoint Questions

**Q1.** An antibody has  $K_d = 100$  pM at 310 K. A competing antibody has  $K_d = 10$  nM. What is the difference in binding free energy? What is the ratio of their Boltzmann probabilities (bound/unbound) at equal free concentrations?

*Answer:* For the first antibody:  $\Delta G_1 = 0.616 \times \ln(10^{-10}) = 0.616 \times (-23.0) = -14.2$  kcal/mol. For the second:  $\Delta G_2 = 0.616 \times \ln(10^{-8}) = -11.3$  kcal/mol. Difference:  $\Delta\Delta G = 2.9$  kcal/mol  $\approx 4.7 RT$ . The first antibody's bound/unbound ratio is  $\exp(4.7) \approx 110$  times larger than the second's — a 100-fold change in  $K_d$  corresponds to a  $\sim 100$ -fold change in the Boltzmann probability ratio.

**Q2.** An SPR experiment measures  $K_d = 2.5$   $\mu$ M at 298 K (room temperature). Convert to  $\Delta G$  and state the result in  $RT$  units.

*Answer:* At 298 K,  $RT = 0.593$  kcal/mol.  $\Delta G = 0.593 \times \ln(2.5 \times 10^{-6}) = 0.593 \times (-12.9) = -7.65$  kcal/mol. In  $RT$  units:  $-12.9 RT$ . Note that working in  $RT$  units is more general because it doesn't depend on the temperature — the value  $\ln K_d = -12.9$  is the dimensionless energy, independent of the temperature convention.

**Q3.** A student says: “If  $\Delta G = RT \ln K_d$ , then the sign of  $\Delta G$  depends on whether  $K_d < 1$  M or  $K_d > 1$  M. That's arbitrary — depends on units.” How do you respond?

*Answer:* The student is identifying a real conceptual issue. The sign convention reflects the **standard state** — the concentration at which bound and unbound are equiprobable. For solution-phase binding, the standard state is 1 M, so  $K_d = 1$  M corresponds to  $\Delta G^\circ = 0$  (equal bound and unbound populations). This is a convention, but it's a universal one in chemistry and biophysics. All experimental  $K_d$  values are reported relative to this 1 M standard state. When  $K_d < 1$  M (binding tighter than the standard state),  $\Delta G^\circ < 0$  — binding is favorable at 1 M reference concentrations. The convention allows consistent comparison across experiments and systems.

**Q4.** The two-state derivation here uses sigmoid. Lecture 6 will prove that sigmoid violates IIA and cannot be the general selection function. How do we reconcile these?

*Answer:* For a two-state system (bound vs. unbound with one specific target), sigmoid is correct — it is the same as softmax applied to two scores. The IIA violation arises when we ask about **selection among multiple competing targets**. Given antigens  $\{1, 2, 3\}$ , the probability of antigen 1 binding depends on antigens 2 and 3 as well. Sigmoid treats each pair independently:  $P(\text{bind } 1) = \sigma(s_1)$ , ignoring  $s_2$  and  $s_3$ . This independence violates IIA. Softmax correctly accounts for the full competitor pool:  $P(\text{bind } 1) = \exp(s_1) / (\exp(s_1) + \exp(s_2) + \exp(s_3))$ . The two-state case is degenerate — there's only one alternative (unbound), so IIA is trivially satisfied. The multi-candidate case is where softmax becomes necessary.

### 14. Connection to Future Lectures

- **Lecture 4, WB3:** The general multi-candidate selection problem — one agent,  $M$  competitor targets — is derived next, yielding the softmax form of competitive binding.

- **Lecture 5:** The energy  $\Delta G$  in this derivation decomposes as a dot product between agent and target features. For nucleic acids (given bilinearity), the decomposition is exact. For proteins (emergent bilinearity), it must be learned.
- **Lecture 6:** The convergence equation — the softmax/Boltzmann distribution — is proven unique. The two-state sigmoid case is shown to be the  $M=2$  special case of the general theorem.
- **Lecture 10 (future):** The affinity calibration framework  $\Delta G = \mathbf{w}s + b$  provides the unit conversion from SFM contrastive scores to absolute  $K_d$  values. The linear form follows directly from the identity  $s = -\Delta G / RT + C$  derived here and in WB3.